

11. A student was asked to find the percentage of calcium carbonate in a sample of chalk. He used the following chemicals.

- Three chalk pieces of identical composition and mass 2.54 g
- Hydrochloric acid solution of concentration 1.00 mol dm^{-3}
- Sodium hydroxide solution of concentration $0.100 \text{ mol dm}^{-3}$

Method

- Use a burette to measure 50.00 cm^3 of hydrochloric acid (an excess) into a 100 cm^3 beaker.
- Put a piece of chalk into the beaker and leave until the reaction finishes.
- Filter the solution into a conical flask to remove any solid impurities.
- Add a few drops of indicator to the solution in the conical flask and titrate against the sodium hydroxide solution.
- Repeat the procedure using the other chalk pieces and calculate a mean titre.
- Use the mean titre to calculate the percentage of calcium carbonate in the chalk sample.

Results

Mass of each chalk piece = 2.54 g

Titration	1	2	3
Final reading / cm^3	16.80	33.05	16.70
Initial reading / cm^3	0.20	16.80	0.35
Titre / cm^3	16.60	16.25	16.35

$$\text{Mean titre} = \frac{16.60 + 16.25 + 16.35}{3} = 16.40 \text{ cm}^3$$

(a) State how the student would know that the reaction between the chalk and acid had finished. [1]

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(b) Suggest and explain **two** improvements to the student's method. [4]

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(c) The equation for the reaction between calcium carbonate and hydrochloric acid is as follows.



Use this and the student's results, including the mean titre of 16.40 cm^3 , to calculate the percentage of calcium carbonate in the chalk sample. [4]

Percentage = %



(d) Comment on the validity of the mean titre calculated.

[1]

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(e) Another student follows the same procedure but filters the solution before the reaction is complete. State what effect, if any, this would have on the value of the titre. Justify your answer.

[2]

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